Quantyx Pricer

Business-friendly platform overview

A non-technical presentation of the pricing platform, its product coverage, architecture, and why it and why it matters for portfolio teams, structuring desks, financial centers and product control. control.

One platform for fixed-income and structured products

Core idea

QuantyxPricer is a QuantLib-based pricing pricing workspace that routes each instrument instrument to the right model using a simple simple model field in its JSON definition.

Technology stack

The platform combines Python, QuantLib, QuantLib, FastAPI, React, and MySQL, giving it giving it both pricing-engine depth and operational usability.

How it is used

It prices instruments, reads benchmark curves, stores asset definitions, and exposes pricing and data-management workflows through REST APIs and a frontend.

Scale today

The documentation describes 17 priceable priceable instrument types across 12 model model files and a backend with more than 20 than 20 endpoints.

What kinds of products can it handle?

Rate-based bonds

Fixed, floating, callable, and investment-grade bonds

Credit products

CLNs plus AT1 or CoCo instruments with extension risk and capital trigger metrics.

Money-market

Discount notes, repos, commercial paper, and money market instruments.

Structured credit

PIK bonds, ABS or MBS tranches, CLO tranches, and leveraged loans.

Equity-linked notes

Barrier convertibles, discount certificates, convertibles, and autocallables.

Inflation and index linked

Government linkers, index-linked notes, and capital-protected structures

How it works

Simple architecture, clear workflow

1

Asset JSON

Each instrument is stored as a JSON definition, usually one per ISIN.

2

Curves

Benchmark discount curves are read from the swap curve catalogue.

3

Dispatcher

The root pricer routes the request based on the model field.

4

API layer

FastAPI exposes pricing, data, and model-management endpoints.

5

Frontend

React provides portfolio views, editing screens, and settings pages.

Data layer

Connected to market data and persistence

External providers

- Refinitiv Datastream for CDS spread histories used in CLN calibration.
- CBonds for bond reference data and prices across fixed-income assets.
- EODHD for equity price histories that support structured-product underlyings.

Internal storage

- MySQL stores assets, prices, and model schemas as JSON payloads.
- Stored procedures handle writes and updates.
- The API layer becomes the operational bridge between pricing logic and user workflows.

Why it matters

The business value in plain language

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Broader coverage

Supports a wide set of products in one framework, from simple bonds to autocallables and securitized assets.

API

Operational usability

Can be integrated into workflows, dashboards, and downstream reporting rather than staying as a desktop-only calculator.

JSON

Scalable setup

Standardized inputs and reusable schemas make the platform easier to extend, test, and govern over time.